

Akshat Raj

+91 9667474395 | akshat.raj1008@gmail.com | [LinkedIn](#) | [Github](#)

Summary

Machine Learning Developer with hands-on experience building data-driven models, computer vision systems, and LLM-powered multi-agent applications. Skilled in Python, TensorFlow, LangChain, and scikit-learn, with a strong foundation in data preprocessing, feature engineering, model training, evaluation, and deployment. Experienced integrating machine learning and generative AI solutions into real-world applications.

Technical Skills

Programming Languages: Python

AI / LLM Frameworks: LangChain, Groq API, Multi-Agent Systems, RAG

Machine Learning: TensorFlow, Keras, Scikit-learn, CNN, LSTM

Computer Vision: OpenCV, MediaPipe, Image Processing

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: GitHub, Streamlit, Jupyter Notebooks

Education

VIT Bhopal University (Expected May 2026)

B.Tech in Computer Science and Engineering in Cloud Computing And Automation

Bhopal, Madhya Pradesh

CGPA: 8.12

Projects

Real-time Sign Language Recognition System | *Python, TensorFlow, OpenCV, MediaPipe* April 2024

- Designed and trained a custom LSTM neural network on 5,200+ images across 26 gesture classes, achieving 96–100% classification accuracy on real-time webcam input.
- Built a real-time data pipeline using OpenCV and MediaPipe to track 21 hand keypoints per frame at 30+ FPS, enabling sub-100ms end-to-end inference latency.
- Deployed an interactive web demo using Streamlit with on-screen confidence score visualization and data augmentation (flip, rotation, brightness jitter) for improved generalization.

Brain Tumor Detection via MRI Classification | *Python, TensorFlow, OpenCV* March 2025

- Developed a CNN classifier for brain MRI scans across 4 categories (Glioma, Meningioma, Pituitary, No Tumor), achieving 93.18% test accuracy.
- Engineered an automated preprocessing pipeline using OpenCV contour detection to isolate brain tissue, reducing background noise and standardizing 512×512 inputs.
- Implemented Keras ImageDataGenerator augmentation and Early Stopping + Model Checkpointing callbacks; built an interactive Streamlit diagnostic interface for instant classification with confidence scores.

Multi-Agent AI Research Assistant | *Python, LangChain, Groq (Llama-3.3-70B), Streamlit* June 2025

- Architected a 4-agent research pipeline (Search, Reader, Writer, Critic) using LangChain and Groq's Llama-3.3-70B to autonomously research a topic and generate structured, source-cited reports.
- Built custom LangChain tools integrating the Tavily Search API and BeautifulSoup-based web scraping to retrieve and parse real-time content from top search results.
- Implemented a critic agent that scores generated reports and returns structured feedback (strengths, gaps, verdict), enabling automated quality review without human input.
- Deployed an interactive Streamlit app with real-time pipeline status tracking, tabbed report/critique/raw-research views, and one-click Markdown export.

Certifications

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI
- Intro to Machine Learning – Kaggle
- Introduction to Artificial Intelligence – Infosys